

POPRAD-GANOVCE, Slovakia

WMO: 119520

Lat: 49.033N

Lon: 20.317E

Elev: 2306

StdP: 13.51

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.4	7.8	-3.3	5.0	4.2	2.3	6.7	10.3	27.8	29.7	25.0	31.9	5.9	40	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	83.9	66.4	80.3	64.9	77.4	63.6	67.9	80.0	66.5	78.0	65.0	75.3	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	96.9	73.0	62.3	91.6	71.3	61.0	87.3	69.5	33.6	80.0	32.4	78.7	31.2	75.3	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.4	20.6	17.5	-3.0	88.3	6.7	2.9	-7.8	90.4	-11.8	92.2	-15.5	93.8	-20.4	95.9		
			-3.6	70.5	6.6	1.3	-8.4	71.4	-12.2	72.2	-15.9	72.9	-20.7	73.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	25.5	28.5	36.8	46.8	53.6	60.9	63.4	63.6	55.4	46.1	38.5	28.4	(6)
(7)		DBStd	15.33	8.63	9.83	8.44	7.52	6.34	5.98	5.58	6.27	6.36	7.32	7.95	7.91	(7)
(8)		HDD50	3172	760	602	412	147	38	0	0	1	21	171	349	671	(8)
(9)		HDD65	7185	1225	1022	875	546	354	153	96	104	293	587	794	1136	(9)
(10)		CDD50	1603	0	0	2	51	149	326	417	421	183	49	5	0	(10)
(11)		CDD65	142	0	0	0	0	0	29	48	60	5	0	0	0	(11)
(12)		CDH74	1522	0	0	0	11	26	316	459	648	62	0	0	0	(12)
(13)		CDH80	327	0	0	0	0	1	58	92	169	7	0	0	0	(13)
(14)	Wind	WSAvg	7.1	7.4	7.5	8.4	7.9	7.7	6.8	6.2	5.9	6.5	6.6	6.9	7.1	(14)
(15)	Precipitation	PrecAvg	23.7	0.9	1.0	1.1	1.7	2.9	3.5	3.3	2.9	2.0	1.7	1.7	1.1	(15)
(16)		PrecMax	39.3	3.0	3.3	3.4	4.6	6.2	8.4	8.7	7.3	4.7	7.4	5.3	3.3	(16)
(17)		PrecMin	16.3	0.0	0.1	0.1	0.2	0.2	0.9	0.3	0.4	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	4.6	0.6	0.7	0.7	1.0	1.3	1.4	1.8	1.5	1.1	1.5	1.1	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	51.1	63.4	75.7	77.0	85.2	86.8	88.9	79.5	70.0	58.4	48.2	(19)
MCWB			42.0	41.4	47.0	55.5	59.5	67.0	67.7	66.1	62.8	56.6	50.7	43.2	(20)	
2%		DB	41.7	46.3	58.5	69.6	73.4	81.3	82.5	84.9	75.2	66.4	54.8	44.7	(21)	
		MCWB	39.5	40.2	46.2	54.4	58.8	65.3	66.2	66.1	61.5	56.0	48.8	41.1	(22)	
5%		DB	39.2	43.1	53.7	65.3	70.3	77.6	78.9	80.7	71.8	62.6	52.0	41.7	(23)	
		MCWB	36.7	38.4	43.7	51.5	57.0	63.6	65.0	65.4	60.1	54.4	47.4	39.0	(24)	
10%		DB	36.8	40.5	49.7	61.4	66.7	73.9	75.8	77.2	67.6	58.1	49.3	38.7	(25)	
		MCWB	34.6	36.6	42.0	49.7	55.7	62.1	63.4	64.2	58.0	52.1	46.0	36.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	43.8	49.0	58.0	61.9	69.9	69.5	69.2	65.0	58.7	52.1	44.1	(27)
MCD B			44.5	48.1	58.9	70.1	72.0	81.3	81.3	81.3	75.7	66.8	56.0	47.4	(28)	
2%		WB	40.1	41.1	46.8	55.1	60.0	67.1	67.7	67.7	62.8	56.7	50.0	41.7	(29)	
		MCD B	41.3	45.2	56.0	68.4	69.9	78.5	79.2	80.4	72.8	64.3	53.4	44.4	(30)	
5%		WB	37.1	39.2	44.8	53.0	58.3	64.6	66.3	66.4	61.1	54.7	48.1	39.2	(31)	
		MCD B	38.8	42.3	52.3	63.6	67.9	75.0	77.2	78.5	70.5	61.4	51.2	41.5	(32)	
10%		WB	34.7	37.1	42.5	50.7	56.6	62.9	64.8	64.8	58.9	52.7	46.1	36.6	(33)	
		MCD B	36.5	40.1	49.2	59.3	65.5	72.0	73.9	75.9	66.0	58.2	49.4	38.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.9	13.4	16.8	18.9	19.3	19.1	20.2	21.9	18.7	16.7	11.5	11.1	(35)
MCDBR			11.1	16.3	24.4	26.2	25.5	25.4	25.2	26.7	24.6	23.5	15.3	12.9	(36)	
MCWBR			8.7	11.3	14.4	13.1	12.7	12.1	11.8	11.4	12.5	13.9	10.4	10.0	(37)	
5% WB		MCDBR	10.2	13.8	22.2	22.9	22.6	22.9	23.1	24.5	21.3	21.8	12.8	12.4	(38)	
		MCWBR	8.7	10.1	13.3	12.0	12.0	11.5	11.2	11.3	11.0	13.4	9.6	10.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.315	0.361	0.421	0.415	0.417	0.432	0.428	0.394	0.359	0.328	0.288	(40)	
taud		2.285	2.227	2.194	2.138	2.222	2.252	2.214	2.232	2.292	2.362	2.376	2.344	(41)		
Ebn at Noon		242	261	266	261	268	268	262	257	253	243	225	226	(42)		
Edh at Noon		23	31	38	45	43	42	43	40	34	27	21	19	(43)		
(44)	All-Sky Solar Radiation	RadAvg	292	532	911	1316	1589	1699	1645	1517	1071	653	335	222	(44)	
(45)		RadStd	41	81	127	143	194	142	177	128	148	91	44	25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+1.03	N/A	N/A	+1.71	+0.90	+0.81	-227	-324	+158	+70

Nomenclature: See separate page